

Final Project - Part 1 of 3

35 Points Possible

4/30/2026

Attempt 1



In Progress

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Unlimited Attempts Allowed

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COE 311K Final Project - Part 1 of 3

Numerical Solution of Second-Order ODEs

Overview

In this project, you will apply numerical methods to solve a real-world engineering problem modeled by a second-order ordinary differential equation (ODE). You will implement two numerical solution methods (Euler's Forward and RK4), analyze their performance, and conduct a stability analysis.

Special Notes on the Final Project

This can be either a group project or an individual project. But we require only one submission. In the header section of your notebook, list all group members - name and EID. If this is an individual project, you will be given 3 bonus points. However, academic integrity rules apply.

Learning Objectives

By completing this project, you will:

- Connect theoretical ODE concepts to real-world engineering applications
- Implement and compare numerical solution methods
- Analyze stability and accuracy of numerical schemes
- Develop technical communication skills through a structured Jupyter Notebook

Project Requirements

1. Model Selection

Choose **ONE** of the following real-world applications:

Option A: Spring-Mass System (Mechanical Oscillations)

$$m \frac{d^2 y}{dt^2} + c \frac{dy}{dt} + ky = 0$$

Where:

- m = mass (kg)
- c = damping coefficient (N·s/m)
- k = spring constant (N/m)
- $y(t)$ = displacement from equilibrium (m)

Example applications: Car suspension systems, building vibrations, mechanical oscillators

Option B: RLC Electrical Circuit

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = V(t)$$

Where:

- L = inductance (H)
- R = resistance (Ω)
- C = capacitance (F)
- $q(t)$ = charge on capacitor (C)
- $V(t)$ = external voltage (V)

Example applications: Audio filters, signal processing circuits, oscillators

Option C: Damped Mechanical System

$$\frac{d^2 y}{dt^2} + 2\zeta\omega \frac{dy}{dt} + \omega^2 y = 0$$

Where:

- ζ = damping ratio (dimensionless)
- ω = natural frequency (rad/s)
- $y(t)$ = displacement or response

Example applications: Earthquake response of structures, shock absorbers, door closers

Option D: Population Dynamics

$$\frac{d^2 P}{dt^2} = rP - kP^2$$

Where:

- $P(t)$ = population size
- r = growth rate
- K = carrying capacity factor

Example applications: Predator-prey systems, ecological modeling, resource management

2. Parameter Research

You must:

- **Research realistic parameter values** from engineering handbooks, scientific literature, or manufacturer specifications, you may use AI as well, but verify and cite where the AI got its information from
- **Cite your sources** for all parameter values
- **Justify your choices** based on the specific real-world scenario you're modeling
- **Define appropriate initial conditions** consistent with your scenario

Example: If modeling a car suspension, specify the make/model, cite the vehicle mass, research typical spring constants for that vehicle class, and choose initial conditions representing hitting a speed bump.

3. Numerical Implementation

Convert to First-Order System

Your second-order ODE must be converted to a system of two first-order ODEs:

For example, if $\frac{d^2y}{dt^2} = f(t, y, \frac{dy}{dt})$, let:

- $y_1 = y$
- $y_2 = \frac{dy}{dt}$

Then:

$$\frac{dy_1}{dt} = y_2$$

$$\frac{dy_2}{dt} = f(t, y_1, y_2)$$

Implement Two Methods

Euler's Forward Method:

For each time step:

$$y_1(t + h) = y_1(t) + h * y_2(t)$$

$$y_2(t + h) = y_2(t) + h * f(t, y_1(t), y_2(t))$$

Fourth-Order Runge-Kutta (RK4):

For each time step, calculate four slopes (k_1, k_2, k_3, k_4)

and combine them with weights: $(k_1 + 2k_2 + 2k_3 + k_4)/6$

Requirements:

- Implement both methods in separate, well-documented functions
- Test with **at least three different step sizes** (e.g., $h = 0.1, 0.05, 0.01$)
- Simulate for a physically meaningful time interval

4. Stability Analysis

Your analysis must include:

a) Step Size Investigation

- Demonstrate how solution accuracy changes with step size
- Identify the **maximum stable step size** for each method
- Show what happens when the step size is **too large** (unstable behavior)

b) Error Quantification

- If an analytical solution exists, calculate and plot the error
- Create a **log-log plot** of error vs. step size to verify convergence rates:
 - Euler's method: $O(h) \rightarrow \text{slope} \approx 1$
 - RK4: $O(h^4) \rightarrow \text{slope} \approx 4$
- If no analytical solution exists, use the finest step size solution as a reference

c) Physical Interpretation

- Discuss whether numerical instabilities lead to **non-physical results**
 - For conservative systems: Does total energy increase artificially?
 - For damped systems: Does the solution grow instead of decay?
- Relate stability to the **physical behavior** of your system

Deliverable: Jupyter Notebook

Submit a **single Jupyter Notebook** to your **GitHub** (.ipynb file) and **submit a PDF** to **Canvas** with the following structure:

Section 1: Introduction & Model Selection (~0.5 pages)

- Brief description of your chosen real-world system
- The governing second-order ODE
- Conversion to a system of first-order ODEs
- Physical meaning of each variable and parameter

Section 2: Parameter Research & Justification (~0.5 pages)

- Table of all parameters with values, units, and sources
- Initial conditions with justification
- Description of the physical scenario being modeled

Section 3: Numerical Methods Implementation (~0.25 pages + code)

- Well-commented Python functions for:
 - Euler's Forward method
 - RK4 method
- Brief explanation of your implementation approach

Section 4: Solutions & Comparison (~0.5 pages + plots)

- Results from both methods with multiple step sizes
- Comparison plots (position vs. time, velocity vs. time)
- Phase portraits (if applicable)
- Discussion of visual differences between methods

Section 5: Stability Analysis (~0.5 pages + plots)

- Demonstration of stability limits for each method
- Error analysis with appropriate plots
- Discussion of stable vs. unstable behavior
- Physical interpretation of results

Section 6: Conclusions (~0.25 pages)

- Which method performed better for your problem?
- Computational cost vs. accuracy tradeoff
- Practical recommendations
- Lessons learned

Formatting Guidelines

Length

- **Total length:** 5-7 pages when printed (including code and plots)
- **Text/markdown:** ~2 pages
- **Code:** ~1-2 pages
- **Plots:** ~2-3 pages (4-6 key figures)

Code Requirements

- Use clear variable names
- Include comments explaining key steps
- Use modular functions (avoid copy-paste code)
- Follow Python best practices

Plot Requirements

- All plots must have:
 - Descriptive titles
 - Labeled axes with units
 - Legends (when multiple curves are shown)
 - Readable font sizes
- Use `matplotlib` or similar plotting library

Citations

- Use any standard citation format (IEEE, APA, or inline links)

Grading Rubric (35 points)

Model & Parameters (10 points)

- Appropriate model selection (4 pts)
- Well-researched, realistic parameters (4 pts)
- Proper citations and justification (2 pts)

Code Implementation (10 points)

- Correct Euler's Forward method implementation (4 pts)
- Correct RK4 implementation (4 pts)
- Code quality: comments, modularity, style (2 pts)

Stability Analysis (10 points)

- Correct step size experiments (4 pts)
- Clear demonstration of stability limits (4 pts)

- Meaningful error analysis with appropriate plots (2 pts)

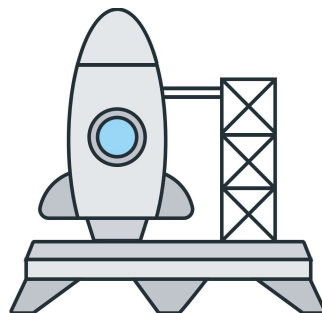
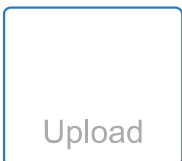
Format (5 points)

- Clear, well-labeled plots with captions (2 pts)
- Concise, well-written explanations (2 pts)
- Overall organization and flow (1 pts)

Submission Instructions

- **Due Date:** 04/30/2026
- **Submit via GitHub:** Upload your .ipynb file
- **Submit via Canvas:** Upload your .pdf of your completed notebook
- **File naming:** LastName_FirstName_COE311K_FinalProject.ipynb/LastName_FirstName_COE311K_FinalProject.pdf
- **If this is a group project:** We only need one submission per group. In the header comments, identify each member (name and EID)
- **Ensure your notebook runs completely:** Before submitting, restart the kernel and run all cells to verify there are no errors

Choose a submission type



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